SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Compare 2 and 4 stroke engines.

Q.22 What do you mean by efficiency, what are the factors affecting it?

Q.23 Describe different propellers used in airplanes.

Q.24 What do you mean by Thrust/power augmentation?

Q.25 Why the propeller blades are twisted?

Q.26 Describe ignition system of engine?

Q.27 What are the various contaminating substances in fuel?

Q.28 What are the different factors affecting engine performance?

Q.29 Describe the operation of fuel content/flow gauges?

Q.30 What are the common characteristics of aviation fuel?

Q.31 Name variousengine Instruments and their use.

Q.32 What are the coolingmethodsin aircraft piston engine?

Q.33 How the rpm of the engine is measured?

Q.34 What are the causes of faults in engines?

Q.35 Explain the various thermometers used.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the working of engine with two and four stroke system with the help of a diagram.

Q.37 Describe fuel system for reciprocating engine. Explain the various types of engine starters?

Q.38 Define the performance parameters of an engine. What are the factors affecting engine performance? How each parameter is controlled?

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4th Sem / Aircraft Maintenance
Subject:- Aircraft Reciprocating Engine

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 An aircraft engine generates________power

a) Electrical b) Mechanical
c) Thermal d) Hydraulic

Q.2 The top dead center is the position of piston when it forms ________ volume in cylinder and bottom dead center is the position of piston when it forms _____ volume in cylinder.

a) Largest, smallest b) Smallest, largest
c) Equal, equal d) None of the above

Q.3 Which of the following is the wrong statement?

a) In the exhaust, retarded timing causes burning of the hydrocarbons
b) The retarded timing improves fuel economy
c) The retarded timing requires the small opening for correct burning of the fuel
d) The exhaust gas temperature becomes higher due to retarded time

Q.4 During a maintenance check, a technician finds that the fire detection system in the engine compartment is malfunctioning. What is the immediate implication of this issue?

a) The engine will consume more fuel.

b) The engine may overheat more frequently.

c) The aircraft will not meet safety regulations and could be at risk of undetected engine fires.

d) The engine will produce more noise.

Q.5 What is the primary purpose of a propeller on an aircraft?

a) To stabilize the aircraft

b) To generate thrust

c) To provide electrical power to the aircraft systems

d) To control the aircraft's altitude

Q.6 The diameter of the piston is called_______

a) Piston b) Bore

c) Stroke d) None of the above

Q.7 Which of the following statement is true?

a) in S.I. engines, combustion of air-fuel mixture is initiated by spark plug

b) in C.I. engines, combustion of air-fuel mixture is self-ignited

c) both of the mentioned

d) none of the mentioned

Q.8 What is the primary purpose of a supercharger in an aircraft engine?

a) To decrease fuel consumption

b) To increase the engine's power output by compressing the intake air

c) To reduce engine noise

d) To cool the engine

Q.9 A pilot reports that the engine exhibits a significant drop in power during takeoff. The supercharger is suspected to be malfunctioning. Which specific supercharger-related component should be inspected first?

a) The oil filter b) The fuel injector

c) The drive belt or gear mechanism

d) The spark plugs

Q.10 Why is it important to monitor the oil pressure gauge during flight?

a) To ensure the aircraft is maintaining a constant speed

b) To verify the oil is being properly circulated and preventing engine wear

c) To check the level of fuel in the tanks

d) To adjust the mixture of air and fuel

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Which type of IC engines are used in aircrafts?

Q.12 Why the pitch of the propellers is varied?

Q.13 For which control, collective pitch is used?

Q.14 What type of fuel is used in aircraft engines?

Q.15 How contamination is checked?

Q.16 What is power augmentation?

Q.17 What are the indicators used for fuelparameters?

Q.18 What is rigging?

Q.19 What instrument is used for fire detection?

Q.20 What is the most common fault in the engine?